ClimateScope Dashboard - Design Documentation

1. Objective

The ClimateScope Dashboard is designed to provide an interactive platform for analyzing global climate data. It helps users understand weather patterns, detect anomalies, and make data-driven decisions using visual analytics.

2. Dashboard Layout

The dashboard is divided into two main sections:

A. Top Control Panel
- Filters: Country/Region, Year, Month, Search

B. Main Dashboard Area
- KPI Cards, Charts, Maps, Alerts

3. Components and Graph Placement

3.1 Live Weather Overview Card

Location: Top-left
Graph Type: Info Card
Details: Temperature, Humidity, Wind, AQI

3.2 Global Temperature Distribution

Location: Center top
Graph Type: Heatmap Map
Details: Shows temperature globally using color gradient

3.3 Top 10 Hottest Countries

Location: Top-right
Graph Type: Bar Chart
Details: Country vs Temperature comparison

3.4 KPI Cards

Location: Middle-left
Graph Type: KPI Cards
Details: Avg Temperature, Humidity, Wind

3.5 Highest Wind Speed
Location: Middle-right
Graph Type: Highlight Card
Details: Shows max wind speed country

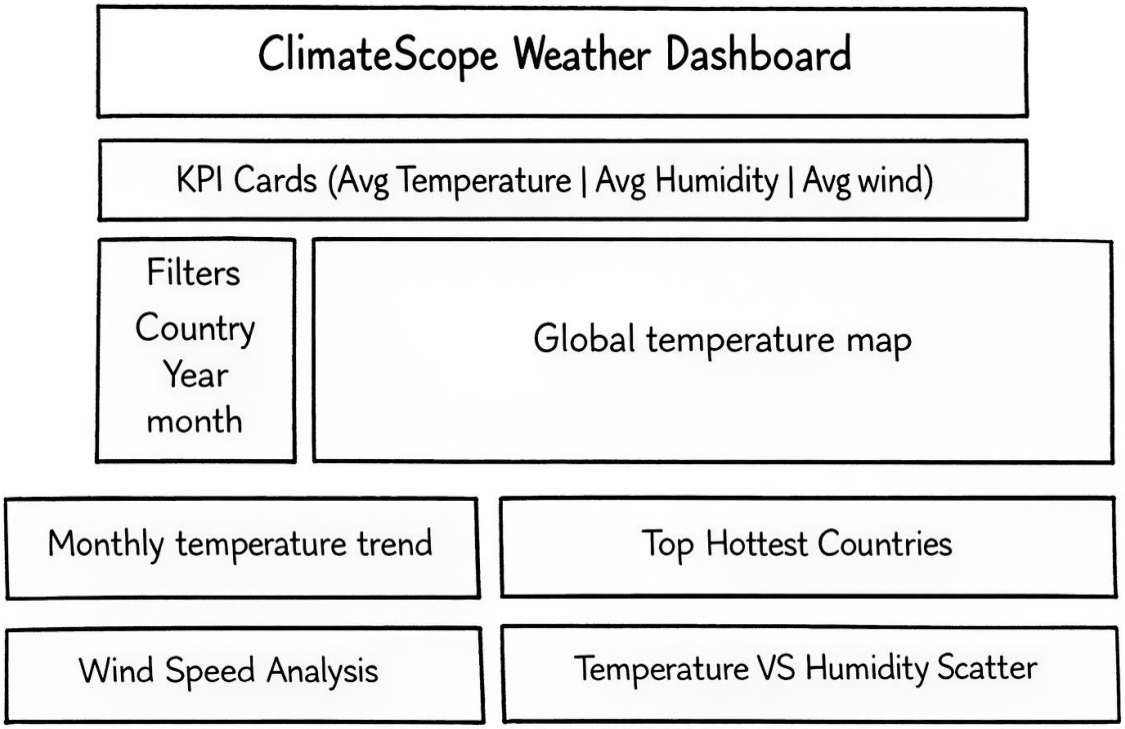
3.6 Monthly Temperature Trend
Location: Bottom-left
Graph Type: Line Chart
Details: Monthly variation of temperature, humidity, wind

3.7 Temperature vs Humidity
Location: Bottom-center
Graph Type: Scatter Plot
Details: Correlation analysis

3.8 Recent Climate Events
Location: Bottom-right
Graph Type: Event List
Details: Heatwaves, rainfall alerts

4. Features
Interactive filtering
Real-time updates
Multiple graphs
User-friendly UI

7. Future Enhancements
AI predictions
Mobile support
API integration
Download reports


ClimateScope Weather Dashboard
KPI Cards (Avg Temperature | Avg Humidity | Avg wind)
Filters
Country
Year
month
Global temperature map
Monthly temperature trend
Top Hottest Countries
Wind Speed Analysis
Temperature VS Humidity Scatter